LAB RECORD

Summer Internship-2025

TOPIC: Learning Bash Script Basics

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26th May 2025-21st July 2025

CONTENTS

WEEK-1

Monday-26th May 2025

a. Setting up a GitHub account

Created a GitHub account to maintain a Lab record throughout the internship period. Created a repository GayathriSP24/Summer-Internship-2025: Summer Internship-2025 May 26- July 21 to enter daily records.

b. Setting up the Linux Platform

Tried setting up MobaXterm, which seemed to have trouble functioning on the PC. Later, with a friend's help, I was able to set up WSL and Ubuntu, saving the day. Did ran some random commands to see if it is working. So, at the end of the day, I have learnt that WSL is already in our Windows 10 and 11 PC, but by turning it on and installing Ubuntu you can run a Linux terminal on a Windows PC.

Tuesday-27th May 2025

a. Working on PPT

After reaching the lab at 9:30, I started working on the presentation for the chapter Bash basics, while trying out every command I came across. Since I needed to brush up on the first 2 chapters before moving to the 3rd Chapter, I decided to revise while making the PPT. Got to know a lot regarding the format of each command to run on a terminal, which I started to record in a Word document. Also, after talking with a friend in the lab, I have decided to try working in VS Code and Notepad to take down. At the end of the day, I have completed the presentation.

b. Updating the Repository

After uploading yesterday's lab record in a Word document last night, it was today that I realised that you can't update the Word document on the platform. So, I converted it to a PDF and uploaded it.

Wednesday-28th May 2025

a. Working on PPT

After reaching the lab at 9:30, I started working on the presentation for the chapter Command-line editing, the second chapter of the book, while brushing up on the topics. I was able to complete up to 2.4. I learnt that my default is vi mode and how to shift to emacs editing mode. I am currently working on the command keys of Emacs one by one while making the presentation.

b. Setting up and working on VS Code

I have installed VS Code in between and shifted to its terminal while using Notepad to copy commands. While working on VS code, I realised that although better to save the commands, VS code at times intercepts certain control keys, like Ctrl+E, etc.

Thursday-29th May 2025

a. Working on PPT

Today's target was to complete Chapter 2 in terms of reviewing and presentation, and to finish the video for Chapter 1 for submission. After arriving at the lab at 9:45, I began working on the presentation using Ubuntu. By 4:15 pm, I had completed the presentation and revised it.

b. Video recording- Chapter 1

By evening, I worked on recording the video using the presentation for chapter 1, which I submitted via Google Classroom. In the end, I submitted this lab record for the day.

Friday-30th May 2025

a. Learning Chapter 3

After arriving at 9:30, I started studying chapter 3, which is about customising our working environment. I was able to complete topics such as special files, Options, and the alias command. Currently, I have reached the shell variables. Throughout the day, I was able to work more efficiently in VS Code compared to how I was at the beginning of the week.

Saturday-31th May 2025

a. Video recording - Emacs editing mode.

Today I decided to work from my room instead of the lab. I have been working on the video part of chapter 2. I recorded the Emacs editing mode in the terminal for the PPT.

WEEK- 2

Monday-2nd June 2025

a. Working on chapter 3

I clocked in at 9:45 am and started working on studying chapter 3 about customising The environment while working on VS Code. By 5 pm, I was left with 12 pages of the chapter, which will be completed by tomorrow.

Tuesday-3rd June 2025

a. Completed chapter 3

I clocked in at 9:45 am and started working on completing chapter 3. I was able to finish it by 12. Then, I started reading chapter 4.

Wednesday-4th June 2025

a. Working on chapter 4

I clocked in at 9:45 am and started working on chapter 4. Today's target is to complete 1/3rd of the chapter.

Thursday-5th June 2025

a. Working on chapter 4

I clocked in at 10:00 am and started working on chapter 4. Today's target is to complete the next 1/3rd of the chapter, which is up to page 285. I did my first program script for sorting a list of musicians with their number of albums. Since I haven't reached conditional statements, I couldn't do it with them. Hopefully, I will be able to start working on it by next week.

```
gayat@GSP-HP:~$ count=5
sort -nr musician.sh | head -n "${count:-10}"

5       Depeche Mode
3       Simple Minds
2       Split Enz
1       Vivaldi, Antonio
```

b. Recording PPT for chapters 2 and 3

I started working on recording the video for the week, based on chapter 2. Then I started working on the PPT for chapter 3. At the end of the day, I submitted my video, and the lab record was uploaded to the repository.

Friday-6th June 2025

a. Working on chapter 4

Today, due to some personal reasons, I decided to work from my room where I worked on chapter 4.

WEEK- 3

Monday-9th June 2025

a. Completed chapter 4

I clocked in at 10:00 am and started working on completing chapter 4. By noon, I had completed it with notes.

b. Making the PPT for chapter 3

From the afternoon, I started to work on making the presentation for chapter 3 using the notes. By 5, I submitted the Lab record ad clocked out of the lab.

Tuesday-10th June 2025

a. Starting chapter 5

I clocked in at 10:00 a.m. and began working on Chapter 5. This is the chapter where you learn about the basis of writing code using bash scripting. I happened to get a chance to attend a presentation by a senior at noon.

b. Completing the PPT for chapter 3

In the evening, I completed the PPT on chapter 3. In the end, I submitted my Lab record.

Wednesday-11th June 2025

a. Working on chapter 5

I clocked in at 10:00 a.m. and began working on Chapter 5. Today's target is to complete about half of it.

b. Completing the PPT for chapter 4

In the evening, I worked on the PPT for chapter 4. In the end, I submitted my Lab record.

Thursday-12th June 2025

a. Working on chapter 5

Due to some reasons, I clocked in at 11:00 and continued working on chapter 5. By the end of the day, I was left with about 11 pages of the chapter left.

Friday-13th June 2025

a. Completing chapter 5

I started working from 9 am at the library, where I completed chapter 5 by noon.
I started chapter 6 in the evening.

b. Submitting the video for Chapter 3

In the evening, I started to record the video for chapter 3 for this week's submission. At the end of the day, I submitted the lab record, video for the week and chapter 5's notes.

WEEK- 4

Monday-16th June 2025

a. Working on chapter 6

I started working from 10 am at the library, where I worked on completing half of chapter 6. By 6:30 pm, I was left with 27 pages of the chapter, which will be completed by tomorrow.

Tuesday-17th June 2025

a. Completed chapter 6

I started working from 10 am at the library, where I worked on completing half of chapter 6, which I completed by the end of the day.

b. Worked on Chapter 4 PPT

In the evening, I started working on the PPT for chapter 4, which will be used for this week's video. By the end of the day, I submitted the lab record and notes for chapter 6.

Wednesday-18th June 2025

a. Started chapter 7

I started reading chapter 7 based on Input-Output and Command line.

b. Worked on using UCSC Genome Browser

Then Dr. Vijay assigned us, a group of 3, to start working on the UCSC browser for genomes, where we were given a deadline for tomorrow to explore the browser using 5 Human gene regions on chromosomes 1,2,3,4,6 and analyse and understand it. I worked on a T2T- telomere to telomere model.

Thursday and Friday-19th and 20th June 2025

a. Working on the project on Genome analysis

After presenting Prof with our understanding of the browser, he assigned us to make an Excel spreadsheet of our findings with the chromosomes and also to install and explore a Linux-based tool called Bedtools.

Later, on Friday, we presented our findings and he assigned us to work on the basic yet most important tools of Bedtools, such as merge, intersect, closest, map, sort, getfasta, etc, by Tuesday.

Saturday-21st June 2025

a. Locked into completing the textbook

Today I completed chapter 7 and a quarter of chapter 8.

WEEK- 5

Monday-23rd June 2025

a. Completed chapter 8

I started working from 10 am at the library, where I worked on completing chapter 8. By noon, I was able to complete the topics.

b. Submission of video for Chapter 4

I recorded the pending video for chapter 4 and submitted it in the classroom. I also worked on creating another repository for the project work.

c. Working on Bedtools

By evening, I started to work on Bedtools, a Linux-based tool for analysis. By the end of the day, I submitted the lab record and notes for chapter 8.

Tuesday-Friday- 24th to 27th June 2025

a. Working on analysing SCFR in the data

After the meeting with Prof on Tuesday, we were assigned to work on 7 samples of Stop Codon Free Regions (SCFRs) in humans and different primate species. We tried to analyse it using GC content, Repeatable elements, microsatellites, difficult regions, etc. We started to work with the human sample and went by the trial-and-error method of analysis. We had to rename our sample data from the accession number of each chromosome to chromosome number and so on.

By Friday, we found that there were elevated levels of GC in many of the samples, many Retrotransposons/LINES, and good levels of CpG Islands in most cases. When reported, we were asked to work on quantifying the findings to make it plausible and rigid.

WEEK- 6

Monday-30th June 2025

a. Working on completing chapters 9 and 10

I started to skim through chapters 9 and 10 to finish it by noon.

b. Working on Bedtools

From the afternoon, I started to work on the Bedtools and Gene analysis.

Tuesday-1st July 2025

a. Working on analysing SCFR in the data

After discussing our methods and our obstacles, my teammates and I came up with an understanding- the assumed hypothesis is that Chromosomes with high GC microsatellites will have a great chance of having high CpG Islands, which indicates to high chance of methylation. This can be an indication of epigenetic modification. So, we started analysing microsatellites by calculating their average for each chromosome separately and also as a whole, to see if we could find a pattern.
I worked on getting the average of GC, AT, GA and TC microsatellites in chromosomes 6 to 10.

Wednesday-Friday-2nd-4th July 2025

a. Working on analysing SCFR in the data

We started working on analysing the data with RepeatMasker genome data from UCSC, and we found a lot of intersects using Bedtools intersect. With this finding, our professor asked us to try getting plots from the data, which includes bar plot, box plot, histogram, mosaic plot, scatter plot etc. We sorted the data in a way the data just contained transposons and retrotransposons.
We had a theory of the number of transposons, especially LINE-L1 being high in the samples of intersection, and they can be a reason for the SCFRs. This was confirmed by the plots, where almost all chromosomes have high levels of L1, except in chromosome, where SINE had a slightly higher levels noted.
When notified, the professor pointed out that we haven't normalised it and asked as to try doing that and see if the data still showed the same pattern.

Sunday-6th July 2025

a. Recorded Video for chapters 5 and 6

In the evening, I recorded the videos for chapters 5 and 6 and uploaded them. And I uploaded the lab record.

WEEK- 7

Monday-7th July 2025

a. Working on Gene Analysis-Bonobo and Chimpanzee

I started working from 10:30 am at the library, where I started working on intersecting Bonobo and Chimpanzee SCFRs data with the RepeatMasker query file to see any match. I then sorted the data chromosome-wise, filtered out the duplicates and then filtered again to just transposons and retrotransposons, which is later normalised and used for bar plots.

Tuesday-8th July 2025

a. Working on Gene Analysis-Human

I started my day, learning and practising the Shapiro-Wilk test, which helps to see if the data is normally distributed.
After working with Chimpanzee and Bonobo, I got into some confusions, which made me to work again in the human SCFRs sample, which I normalised and plotted using bash and R. After checking the data manually using the repeat count, I derived earlier, I was able to conclude at the end of the day that the data was normalised.

Wednesday-9th July 2025

a. Meeting the Professor

We met our professor with the plots, and we were asked to work on other species, compile all the codes and scripts used, and wrap up by the end of the week.

Thursday-Saturday-10th -12th July 2025

a. Working on Normalising and plotting with Bonobo and Chimpanzee

I started compiling each step into protocols from each document, and normalised the data using bash and plotted using R. All the files are compiled into a Google Drive folder for better accessibility between the team members.